

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284435

Kind Code

A1

Publication Date

September 11, 2025

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APPARATUS WITH CALIBRATION INPUT MECHANISM AND METHODS FOR OPERATING THE SAME

Abstract

Methods, apparatuses, and systems related to calibrating memory circuitry according to externally provided reference voltage are described. A memory device may include a calibration control logic that at least isolates an internal reference voltage from an internal buffer. The internal buffer may receive and process the externally provided reference voltage instead of command-address signals for calibration purposes.

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Family ID: 1000008619761

Appl. No.: 19/215187

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 18513438 20231117 parent-grant-document US 12333191 child US 19215187

us-provisional-application US 63439524 20230117

Publication Classification

Int. Cl.: G06F3/06 (20060101)

U.S. Cl.:

CPC G06F3/0659 (20130101); G06F3/0604 (20130101); G06F3/0679 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/513,438, filed Nov. 17, 2023; which claims priority to U.S. Provisional Patent Application No. 63/439,524, filed Jan. 17, 2023, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The disclosed embodiments relate to devices, and, in particular, to semiconductor memory devices with a calibration input mechanism and methods for operating the same.

BACKGROUND

[0003] An apparatus (e.g., a processor, a memory system, and/or other electronic apparatus) can include one or more semiconductor circuits configured to store and/or process information. For example, the apparatus can include a memory device, such as a volatile memory device, a non-volatile memory device, or a combination device. Memory devices, such as dynamic random-access memory (DRAM), can utilize electrical energy to store and access data.

[0004] With technological advancements in other areas and increasing applications, the market is continuously looking for faster, more efficient, and smaller devices. In addition, the fabricated devices are used to support growing number of new devices and new implementation modes. To meet the market demand, the semiconductor devices are being pushed to the limit with various improvements. Improving devices, generally, may include increasing circuit density, increasing operating speeds or otherwise reducing operational latency, increasing reliability, increasing data retention, reducing power consumption, or reducing manufacturing costs, among other metrics. However, such improvements can often introduce challenges, such as in accommodating testing/validation functions with normal operational functions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a block diagram of an apparatus in accordance with an embodiment of the present technology.

[0006] FIG. 2 is a schematic block diagram of an example mode configuration in accordance with an embodiment of the present technology.

[0007] FIG. 3 is an example timing diagram for the mode configuration in accordance with an embodiment of the present technology.

[0008] FIG. 4 is a flow diagram illustrating an example method of operating an apparatus in accordance with an embodiment of the present technology.

[0009] FIG. 5 is a schematic view of a system that includes an apparatus in accordance with an embodiment of the present technology.

DETAILED DESCRIPTION

[0010] As described in greater detail below, the technology disclosed herein relates to an apparatus, such as for memory systems, systems with memory devices, related methods, etc., for controlling inputs during different operation modes. In some embodiments, a memory device, such as a DRAM, can selectively isolate internal power (e.g., internal reference voltage (VREF)) and use an externally provided voltage (e.g., VREF Force provided by an external tester device) for one or more operations or operating modes (e.g., calibration). For example, the memory device can use the externally provided voltage for calibrating DC offset and/or other settings at command-address (CA) input buffer (IB), such as for meeting speed requirements. The memory device can include an input control circuit between a CA pad and the CA IB. The input control circuit can include a set of

switches configured to select the internal VREF during normal operations and the external VREF Force during test/calibration mode. The input control circuit can select the external VREF Force according to a mode control signal (e.g., an asynchronous calibration enable signal, such as **tmIBCalF**).

[0011] The input control circuit configured to leverage the externally provided voltage during calibration (e.g., CA IB calibration) can allow the apparatus to calibrate for higher communication speeds, such as necessary for newer memory devices (e.g., Double Data Rate 5 (DDR5) devices). The ability to calibrate for and operate at higher speeds can significantly outweigh the additional unknown factor or potential failure introduced by the external power source. Moreover, in comparison to some conventional approaches, the input control circuit can allow the apparatus to avoid the use of an in-line pass gate between the external pad and the CA IB. The input control circuit can further allow the apparatus to avoid the cost of adding separate pads or parallel TTL input buffers as required for some conventional circuits.

Example Environment

[0012] FIG. 1 is a block diagram of an apparatus **100** (e.g., a semiconductor die assembly, including a three-dimensional integration (3DI) device or a die-stacked package) in accordance with an embodiment of the present technology. For example, the apparatus **100** can include a DRAM or a portion thereof that includes one or more dies/chips.

[0013] The apparatus **100** may include an array of memory cells, such as memory array **150**. The memory array **150** may include a plurality of banks (e.g., banks 0-15), and each bank may include a plurality of word-lines (WL), a plurality of bit lines (BL), and a plurality of memory cells arranged at intersections of the word-lines and the bit lines. Memory cells can include any one of a number of different memory media types, including capacitive, magnetoresistive, ferroelectric, phase change, or the like. The selection of a word-line WL may be performed by a row decoder **140**, and the selection of a bit line BL may be performed by a column decoder **145**. Sense amplifiers (SAMP) may be provided for corresponding bit lines BL and connected to at least one respective local I/O line pair (LIOT/B), which may in turn be coupled to at least respective one main I/O line pair (MIOT/B), via transfer gates (TG), which can function as switches. The sense amplifiers and transfer gates may be operated based on control signals from decoder circuitry, which may include the command decoder **115**, the row decoders **140**, the column decoders **145**, any control circuitry of the memory array **150**, or any combination thereof. The memory array **150** may also include plate lines and corresponding circuitry for managing their operation.

[0014] The apparatus **100** may employ a plurality of external terminals that include command and address terminals coupled to a command bus and an address bus to receive command signals (CMD) and address signals (ADDR), respectively. The apparatus **100** may further include a chip select terminal to receive a chip select signal (CS), clock terminals to receive clock signals CK and CKF, data clock terminals to receive data clock signals WCK and WCKF, data terminals DQ, RDQS, DBI, and DMI, power supply terminals VDD, VSS, and VDDQ.

[0015] The command terminals and address terminals may be supplied with an address signal and a bank address signal (not shown in FIG. 1) from outside. The address signal and the bank address signal supplied to the address terminals can be transferred, via a command/address (CA) input circuit **105**, to an address decoder **110**. The address decoder **110** can receive the address signals and supply a decoded row address signal (XADD) to the row decoder **140**, and a decoded column address signal (YADD) to the column decoder **145**. The address decoder **110** can also receive the bank address signal and supply the bank address signal to both the row decoder **140** and the column decoder **145**.

[0016] The apparatus **100** can include an input control circuit **190** configured to select an input voltage, such as between an internal reference voltage (VREF) or an externally provided reference voltage (VREF Force). The input control circuit **190** can select the input voltage according to an operating mode. For example, the input control circuit **190** can select the VREF Force when the

apparatus **100** is operating in a testing/validation mode and/or a calibration mode. The input control circuit **190** can select the VREF Force for calibrating the DC offset for the CA input buffer. In some embodiments, the CA input circuit **105** can include the input control circuit **190**. In other embodiments (not shown), the apparatus **100** can include a separate test/calibration circuit having the input control circuit **190**. Details regarding the input control circuit **190** are described below. [0017] The apparatus **100** can include an IB calibration circuit **191** configured to implement a CA IB calibration process to determine the DC offset using the externally provided VREF Force. For example, the IB calibration circuit **191** can include logic configured to control operations of the apparatus **100** during the CA IB calibration process, including controlling state transitions and corresponding timing/signaling for the process, capturing a CA state of each IB associated with the CA pads, formatting and reporting the CA state to a tester or a system operator, or the like. Also, the IB calibration circuit **191** can include storage circuits, such as latches, fuses, etc. for storing the captured CA states and/or trim settings that represent the determined DC offset. The trim setting can be applied to the corresponding IB, such as during apparatus initialization, thereby adjusting or fine-tuning the IB to perform within expected signal parameters and receive the CA input during normal operation.

[0018] The command and address terminals may be supplied with command signals (CMD), address signals (ADDR), and chip select signals (CS), from a memory controller and/or a nefarious chipset. The command signals may represent various memory commands from the memory controller (e.g., including access commands, which can include read commands and write commands). The chip select signal may be used to select the apparatus **100** to respond to commands and addresses provided to the command and address terminals. When an active chip select signal is provided to the apparatus **100**, the commands and addresses can be decoded, and memory operations can be performed. The command signals may be provided as internal command signals ICMD to a command decoder **115** via the command/address input circuit **105**. The command decoder **115** may include circuits to decode the internal command signals ICMD to generate various internal signals and commands for performing memory operations, for example, a row command signal to select a word-line and a column command signal to select a bit line. The command decoder **115** may further include one or more registers for tracking various counts or values (e.g., counts of refresh commands received by the apparatus **100** or self-refresh operations performed by the apparatus **100**).

[0019] Read data can be read from memory cells in the memory array **150** designated by row address (e.g., address provided with an active command) and column address (e.g., address provided with the read). The read command may be received by the command decoder **115**, which can provide internal commands to input/output circuit **160** so that read data can be output from the data terminals DQ, RDQS, DBI, and DMI via read/write amplifiers **155** and the input/output circuit **160** according to the RDQS clock signals. The read data may be provided at a time defined by read latency information RL that can be programmed in the apparatus **100**, for example, in a mode register (not shown in FIG. 1). The read latency information RL can be defined in terms of clock cycles of the CK clock signal. For example, the read latency information RL can be a number of clock cycles of the CK signal after the read command is received by the apparatus **100** when the associated read data is provided.

[0020] Write data can be supplied to the data terminals DQ, DBI, and DMI according to the WCK and WCKF clock signals. The write command may be received by the command decoder **115**, which can provide internal commands to the input/output circuit **160** so that the write data can be received by data receivers in the input/output circuit **160** and supplied via the input/output circuit **160** and the read/write amplifiers **155** to the memory array **150**. The write data may be written in the memory cell designated by the row address and the column address. The write data may be provided to the data terminals at a time that is defined by write latency WL information. The write latency WL information can be programmed in the apparatus **100**, for example, in the mode

register. The write latency WL information can be defined in terms of clock cycles of the CK clock signal. For example, the write latency information WL can be a number of clock cycles of the CK signal after the write command is received by the apparatus **100** when the associated write data is received.

[0021] The power supply terminals may be supplied with power supply potentials V.sub.DD and V.sub.SS. These power supply potentials V.sub.DD and V.sub.SS can be supplied to an internal voltage generator circuit **170**. The internal voltage generator circuit **170** can generate various internal potentials V.sub.PP, V.sub.OD, V.sub.ARY, V.sub.PERI, and the like based on the power supply potentials V.sub.DD and V.sub.SS. The internal potential V.sub.PP can be used in the row decoder **140**, the internal potentials V.sub.OD and V.sub.ARY can be used in the sense amplifiers included in the memory array **150**, and the internal potential V.sub.PERI can be used in many other circuit blocks.

[0022] The power supply terminal may also be supplied with power supply potential V.sub.DDQ. The power supply potential V.sub.DDQ can be supplied to the input/output circuit **160** together with the power supply potential VSS. The power supply potential V.sub.DDQ can be the same potential as the power supply potential V.sub.SS in an embodiment of the present technology. The power supply potential V.sub.DDQ can be a different potential from the power supply potential V.sub.DD in another embodiment of the present technology. However, the dedicated power supply potential V.sub.DDQ can be used for the input/output circuit **160** so that power supply noise generated by the input/output circuit **160** does not propagate to the other circuit blocks.

[0023] The clock terminals and data clock terminals may be supplied with external clock signals and complementary external clock signals. The external clock signals CK, CKF, WCK, WCKF can be supplied to a clock input circuit **120**. The CK and CKF signals can be complementary, and the WCK and WCKF signals can also be complementary. Complementary clock signals can have opposite clock levels and transition between the opposite clock levels at the same time. For example, when a clock signal is at a low clock level a complementary clock signal is at a high level, and when the clock signal is at a high clock level the complementary clock signal is at a low clock level. Moreover, when the clock signal transitions from the low clock level to the high clock level the complementary clock signal transitions from the high clock level to the low clock level, and when the clock signal transitions from the high clock level to the low clock level the complementary clock signal transitions from the low clock level to the high clock level.

[0024] Input buffers included in the clock input circuit **120** can receive the external clock signals. For example, when enabled by a clock/enable signal from the command decoder **115**, an input buffer can receive the clock/enable signals. The clock input circuit **120** can receive the external clock signals to generate internal clock signals ICLK. The internal clock signals ICLK can be supplied to an internal clock circuit **130**. The internal clock circuit **130** can provide various phase and frequency controlled internal clock signals based on the received internal clock signals ICLK and a clock enable (not shown in FIG. **1**) from the command/address input circuit **105**. For example, the internal clock circuit **130** can include a clock path (not shown in FIG. **1**) that receives the internal clock signal ICLK and provides various clock signals to the command decoder **115**. The internal clock circuit **130** can further provide input/output (IO) clock signals. The IO clock signals can be supplied to the input/output circuit **160** and can be used as timing signals for determining output timing of read data and/or input timing of write data. The IO clock signals can be provided at multiple clock frequencies so that data can be output from and input to the apparatus **100** at different data rates. A higher clock frequency may be desirable when high memory speed is desired. A lower clock frequency may be desirable when lower power consumption is desired. The internal clock signals ICLK can also be supplied to the internal clock circuit **130** and thus various internal clock signals can be generated.

[0025] The apparatus **100** can be connected to any one of a number of electronic devices capable of utilizing memory for the temporary or persistent storage of information, or a component thereof.

For example, a host device of apparatus **100** may be a computing device such as a desktop or portable computer, a server, a hand-held device (e.g., a mobile phone, a tablet, a digital reader, a digital media player), or some component thereof (e.g., a central processing unit, a co-processor, a dedicated memory controller, etc.). The host device may be a networking device (e.g., a switch, a router, etc.) or a recorder of digital images, audio and/or video, a vehicle, an appliance, a toy, or any one of a number of other products. In one embodiment, the host device may be connected directly to apparatus **100**; although in other embodiments, the host device may be indirectly connected to memory device (e.g., over a networked connection or through intermediary devices).
Input Control Circuit

[0026] FIG. **2** is a schematic block diagram of an example mode configuration **200** in accordance with an embodiment of the present technology. In some embodiments, the mode configuration **200** can have a set of the apparatus **100** (e.g., memory devices **100n-1**, **100n**, **100n+1**, and the like) connected to at least one tester **202** via corresponding input connectors **204**. The tester **202** can be configured to test, validate, and/or calibrate functionalities of the connected set of devices by providing inputs and/or controlling one or more operations at the devices. The inputs and the controlled operations can be used to determine whether the devices are operating as expected and/or adjust one or more operational functions, parameters, or the like. The tester **202** can be configured to test or calibrate one or multiple devices (e.g., devices under test (DUTs **100n-1**, **100n**, **100n+1**, and so forth)) simultaneously, such as via parallel connections provided by the input connectors **204**. In one or more embodiments, the mode configuration **200** can be for testing circuits (e.g., the CA input circuit **105** of FIG. **1** and/or other circuits described above) associated with one more CA pads **206** of each DUT.

[0027] The tester **202** can be configured to provide external reference voltage (VREF Force) **212** to each DUT. In some embodiments, the external reference voltage can correspond to half of a voltage supply (e.g., VDD/2). The VREF Force **212** can replace an internal reference voltage (VREFCA(CS)) **214** during background operations, such as device setup, testing, calibration, or the like. In other words, the DUTs can perform one or more operations using the external reference voltage **212** during testing/validation/calibration processes instead of the internal reference voltage **214** that is otherwise utilized under normal operating modes. The tester **202** can provide the external reference voltage **212** through the one or more CA pads **206** of each DUT and the corresponding input connectors **204**.

[0028] As an illustrative example, the DUT and/or the tester **202** can respond to a mode control signal **216** (e.g., an asynchronous calibration enable signal, such as tmIBCalF) associated with the one or more designated processes. As illustrated in FIG. **2**, in response to activation of tmIBCalF, the DUT can implement asynchronous CA IB calibration operations to calibrate input buffers (IBs) **230** therein. In normal operation, the IB **230** can include circuits (e.g., an input buffer) configured to receive the CA signal communicated through the CA pads **206**. To improve the signal receiving capabilities and the signal alignment, the IB **230** can be configured to adjust for individual hardware characteristics of each IB. For example, the IB **230** can be implemented as a differential buffer having one of the inputs connected to the internal reference voltage **214** as a configurable offset DC voltage. The asynchronous calibration process can be used to calibrate or set the offset voltage (via, e.g., trim setting) to a level appropriate for each IB **230** to meet the signal speed requirements.

[0029] For the illustrated embodiments, the DUTs (e.g., the apparatus **100**) can be configured to utilize the external reference voltage **212** from the tester **202** to implement the targeted process (e.g., the IB calibration process). Accordingly, during the process, the CA pads **206** can be used as common-mode input (e.g., instead of internal VREF), and the DUT can enable and operate the internal calibration circuits (not shown) using the external reference voltage **212**. Clocks to the DUT and/or one or more designated circuits can be suspended during asynchronous capture, and the asynchronous input can be used to capture the CA output. The DUT can select and utilize the

appropriate voltage according to the mode control signal **216**.

[0030] For the power input selection, each DUT can include a calibration control logic **220** disposed between the one or more CA pads **206** and the corresponding IB **230**. The calibration control logic **220** can be configured to select offset calibration setting provided to the IB **230**. For example, the calibration control logic **220** can include an internal-isolation switch **222** connected in series with and between the IB **230** and the internal reference voltage **214**. In some embodiments, the calibration control logic **220** can include an control switch (e.g., a differential-connection switch **224**) configured to control the input level provided to the IB **230**. For example, the differential-connection switch **224** can be connected to the external reference voltage **212** (e.g., the one or more CA pads **206**) and in parallel with the differential inputs to the IB **230**. Accordingly, the differential-connection switch **224** can be used to provide related or same power input to the differential inputs to the IB **230**.

[0031] The calibration control logic **220** can be configured to operate according to the mode control signal **216**. For example, the internal-isolation switch **222** and/or the differential-connection switch **224** can be controlled by (via, e.g., the gate connection(s) to) the mode control signal **216**. In some embodiments, the internal-isolation switch **222** and the differential-connection switch **224** can be complementary. For example, such as by using an n-type device and a p-type device or by using a mode signal inverter **226** connected to the internal-isolation switch **222** or the differential-connection switch **224**. Accordingly, when the mode control signal **216** is activated, the calibration control logic **220** can be configured to open the internal-isolation switch **222** and close the differential-connection switch **224**. In response, the calibration control logic **220** can remove the internal reference voltage **214** from the IB **230** and/or connect the external reference voltage **212** to the IB **230** according to the targeted configuration (e.g., in parallel to both differential inputs). When the mode control signal **216** is deactivated (e.g., normal or other non-targeted operating modes), the calibration control logic **220** can be configured to close the internal-isolation switch **222** and open the differential-connection switch **224**.

[0032] Utilizing the external reference voltage **212** via the calibration control logic **220** provides the apparatus **100** with improved accessibility to the voltage read at the IB **230**. The apparatus **100** can analyze and/or output one or more known voltage levels provided by the tester **204** using the actual voltage received at the IB **230**. The voltage received at the IB **230** can be used to set the configurable offset level (e.g., the internal reference voltage **214**) without providing an external connection to the IB **230** within the apparatus **100**. As such, the use of the external reference voltage **212** via the calibration control logic **220** can be leveraged to reduce special-purpose (e.g., calibration only) circuit components, such as probe-only pads or connections for external measuring equipment, parallel pass gates or signal paths, and the like. The reduction in the special-purpose circuit components can lead to reduced circuit footprint, further reducing the manufacturing cost and circuit size for the apparatus **100**. Along with the reduced circuit components and footprint, the calibration control logic **220** and the use of the external reference voltage **212** for IB calibration can reduce or eliminate corresponding capacitances and/or resistances (e.g., resulting from TTL CA IB in parallel with native CA IB or external measuring equipment) that can cause calibration errors.

[0033] The determined/calibrated offset level can be implemented as a trim setting. For example, the offset level can be determined based on iteratively testing different trim settings for the IB **230** with the external reference voltage **212**. The apparatus **100** and/or the tester **202** can determine the trim setting that provided the optimal calibration results (e.g., highest accuracy in detecting the input signal), and the trim setting can be stored (via, e.g., fuse) in the apparatus **100**. The stored trim setting (tmfzIBSel[2:0]) can be applied to the IB **230**, such as during initialization, to normalize and tune the IB **230** for CA communication.

[0034] In implementing the calibration process, an asynchronous pin can be used to trigger the capture of the IB result. The captured IB result can then return to synchronous mode. In some

embodiments, the calibration control logic **220** and the IB **230** can be coupled to an output formatting circuit **240** configured to report the output in serial sequence. For example, the calibration output can be captured in parallel and processed at a calibration management circuit (not shown), such as firmware and/or logic. The captured result (IbResult[n-1]) can be provided in parallel to a multiplexer or a serializer that is activated according to a capture/shift mode signal. The output formatting circuit **240** can include a set of flipflops connected to the serialized output and configured to shift out the output data to a test interface according to a capture/shift out signal (e.g., a sequence of pulses).

[0035] The output formatting circuit **240** can use the serialized output to reduce routing. For example, given the parallel to serial transition, the apparatus **100** can include one path or connection from the IB **230** to the calibration processing circuit instead of a bus of parallel connections for the parallel output.

Timing Diagram

[0036] FIG. **3** is an example timing diagram **300** for the mode configuration in accordance with an embodiment of the present technology. The timing diagram **300** can illustrate a set of signal levels (e.g., voltages applied at a set of asynchronous pins) used to control an IB calibration sequence. In some embodiments, the IB calibration can be implemented based on the mode control signal **216**, an asynchronous signal **302** (e.g., an ASYNC), a test data signal (TDQ) **304**, a CA input **306**, or a combination thereof.

[0037] For the example illustrated in FIG. **3**, the calibration mode configuration and the corresponding process can be initiated when the mode control signal **216** is activated, such as at t1. The asynchronous signal **302** can be high at t1, and in response, the apparatus **100** of FIG. **1** can drive or control the test data signal **304**.

[0038] After a predetermined period, such as at t2, the asynchronous signal **302** can transition, such as to a low level. The transition of the asynchronous signal **302** can be used to allow the tester **202** of FIG. **2** to control of one or more signals (e.g., the test data signal **304**) and/or corresponding calibration steps.

[0039] In response to the transition of the asynchronous signal **302**, at t3, the tester **202** can drive or control the test data signal **304** and provide the external reference voltage **212** of FIG. **2** (common mode VREF) through the CA pad **206** of FIG. **2** (e.g., CA7). The apparatus **100** can measure and process the external reference voltage **212** at the IB **230** of FIG. **2**.

[0040] After a predetermine duration, at t4, the tester **202** can transition the test data signal **304**, such as from low to high. In response, the apparatus **100** can capture the CA state. At t5, the asynchronous signal **302** can be transitioned, such as from low to high, thereby allowing the apparatus **100** to resume control of the test data signal **304**. Afterwards, the apparatus **100** can use the output formatting circuit **240** to communicate the calibration output serially through the test data signal **304**. As illustrated in the timing diagram **300**, a clock signal **308** can be suspended during the IB calibration process.

Control Flow

[0041] FIG. **4** is a flow diagram illustrating an example method **400** of operating an apparatus (e.g., the apparatus **100** of FIG. **1** along with the tester **202** of FIG. **2**) in accordance with an embodiment of the present technology. For example, the method **400** can be for operating the calibration control logic **220** of FIG. **2**, the IB **230** of FIG. **2**, the mBIST or other calibration circuit, the output formatting circuit **240** of FIG. **2**, or a combination thereof, such as to leverage the external reference voltage **212** of FIG. **2** in calibrating the IB **230** connected to the CA pad **206** of FIG. **2**. The method **400** can further correspond to the timing diagram **300** of FIG. **3**.

[0042] At block **402**, the apparatus **100** (via, e.g., firmware, control logic, calibration circuit, or the like) can enter calibration test mode. The apparatus **100** can receive a mode control signal (e.g., the mode control signal **216** of FIG. **2**), such as set by a system operator, the tester **202**, etc. In some embodiments, the apparatus **100** can receive and/or send a sequence of signal settings along with

the mode control signal **216** as illustrated in block **404**. For example, the apparatus **100** can receive the mode control signal **216**, sent the asynchronous signal **302** of FIG. 3, receive the test data signal **304** of FIG. 3 set by the tester **202**, or a combination thereof as illustrated in FIG. 3.

[0043] In response to implementation of the predetermined signal setting/sequence, the apparatus **100** can initiate the calibration circuit/process, including the calibration control logic **220** of FIG. 2. For example, the apparatus **100** can close the differential-connection switch **224** of FIG. 2 and/or open the internal-isolation switch **222** of FIG. 2 in response to the activation of the mode control signal **216**.

[0044] At block **406**, the apparatus **100** can receive the VREF on repurposed connections, such as through the CA pads **206** of FIG. 2. The tester **202** can provide the external reference voltage **212** can provide the common mode VREF as illustrated at t3 in FIG. 3. The tester **202** can provide the external reference voltage **212** (e.g., 30%-70% of VDD) to the CA pads **206** of one or more DUTs simultaneously and in parallel through corresponding input connectors **204** of FIG. 2. In some embodiments, the tester **202** can provide the external reference voltage **212** simultaneously to all CA pads on all connected DUTs. The calibration control logic **220** can route the received external reference voltage **212** from the CA pads **206** to the IB **230**.

[0045] At block **408**, the apparatus **100** can capture the CA outputs in parallel. The apparatus **100** can use the signal transition of the asynchronous signal **302** to initiate the capture, such as illustrated at t4 in FIG. 3. The apparatus **100** can capture the CA outputs based on the status or operation of the IB **230** in response to the received external reference voltage **212**. The apparatus **100** can capture the CA outputs according to a predetermined routine, storage circuitry (e.g., latches), and the like. A subsequent transition of the asynchronous signal **302**, such as the high to low transition shown at the right side of FIG. 3, can return the apparatus back to synchronous mode. Between such transitions (e.g., before the high to low transition), the apparatus **100** and/or the tester **202** can reset and stabilize the CA circuitry back into valid command states. The apparatus **100** and/or the tester **202** can reset and stabilize the CA circuitry according to a predetermined sequence of signals, commands, delays, internal circuit operations at the CA input circuit **105** of FIG. 1 and/or the tester **202**, or the like.

[0046] At block **410**, the apparatus **100** can synchronously read back the results. For example, the apparatus **100** can use the output formatting circuit **240** to convert the parallel output data into a serially shifted output as described above. The apparatus **100** and/or the tester **202** can implement one or more operations to ensure that the CA communication functions are returned to normal operating states, such as after t5 as illustrated in FIG. 3.

[0047] In some embodiments, the tester **202** and the apparatus **100** can perform the calibrating operations described above (e.g., blocks **402** to **410**) using different offset trims. For testing different trims, the method **400** can begin with the offset trim initialized to a predetermined state (e.g., lowest or highest setting), and the apparatus **100** and/or the tester **202** can increment the offset trim setting as illustrated at block **412**. At decision block **414**, the apparatus **100** and/or the tester **202** can determine whether the trim setting (e.g., a representation of the loop counter). When the iterative loop is not at the end and untested trim settings remain, the tester **202** and the apparatus **100** can repeat the processes described above for blocks **402-410** with the newly incremented offset trim (block **412**). When the iterative loop is at the end and all trim settings have been tested, the tester **202** and/or the apparatus **100** can analyze the read results across the trim settings to determine the optimal setting (e.g., setting with lowest read error, fastest response time, or the like). At block **416**, the apparatus **100** can set the offset trim for the IB **230**. The tester **202** and/or the system operator can provide the determined offset trim setting, and the apparatus **100** can store the determined trim setting, such as using fuses or other non-volatile memory. The apparatus **100** can apply and use the offset trim setting for the corresponding IB **230** during normal operation.

[0048] FIG. 5 is a schematic view of a system that includes an apparatus in accordance with embodiments of the present technology. Any one of the foregoing apparatuses (e.g., memory

devices) described above with reference to FIGS. 1-4 can be incorporated into any of a myriad of larger and/or more complex systems, a representative example of which is system **580** shown schematically in FIG. 5. The system **580** can include a memory device **500**, a power source **582**, a driver **584**, a processor **586**, and/or other subsystems or components **588**. The memory device **500** can include features generally similar to those of the apparatus described above with reference to FIGS. 1-4, and can therefore include various features for performing a direct read request from a host device. The resulting system **580** can perform any of a wide variety of functions, such as memory storage, data processing, and/or other suitable functions. Accordingly, representative systems **580** can include, without limitation, hand-held devices (e.g., mobile phones, tablets, digital readers, and digital audio players), computers, vehicles, appliances and other products. Components of the system **580** may be housed in a single unit or distributed over multiple, interconnected units (e.g., through a communications network). The components of the system **580** can also include remote devices and any of a wide variety of computer readable media.

[0049] From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but that various modifications may be made without deviating from the disclosure. In addition, certain aspects of the new technology described in the context of particular embodiments may also be combined or eliminated in other embodiments. Moreover, although advantages associated with certain embodiments of the new technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

[0050] In the illustrated embodiments above, the apparatuses have been described in the context of DRAM devices. Apparatuses configured in accordance with other embodiments of the present technology, however, can include other types of suitable storage media in addition to or in lieu of DRAM devices, such as, devices incorporating NAND-based or NOR-based non-volatile storage media (e.g., NAND flash), magnetic storage media, phase-change storage media, ferroelectric storage media, etc.

[0051] The term “processing” as used herein includes manipulating signals and data, such as writing or programming, reading, erasing, refreshing, adjusting or changing values, calculating results, executing instructions, assembling, transferring, and/or manipulating data structures. The term data structure includes information arranged as bits, words or code-words, blocks, files, input data, system-generated data, such as calculated or generated data, and program data. Further, the term “dynamic” as used herein describes processes, functions, actions or implementation occurring during operation, usage or deployment of a corresponding device, system or embodiment, and after or while running manufacturer's or third-party firmware. The dynamically occurring processes, functions, actions or implementations can occur after or subsequent to design, manufacture, and initial testing, setup or configuration.

[0052] The above embodiments are described in sufficient detail to enable those skilled in the art to make and use the embodiments. A person skilled in the relevant art, however, will understand that the technology may have additional embodiments and that the technology may be practiced without several of the details of the embodiments described above with reference to FIGS. 1-5.

Claims

1. A system, comprising: an input buffer (IB) configured to isolate from a load (1) a command-address (CA) signal during normal operation according to a trim setting and an internal reference voltage and (2) an external reference voltage during a calibration process that determines the trim setting; and a logic coupled to the IB and configured to operate, during the calibration process, to (1) provide the external reference voltage to the IB and (2) isolate the IB from the internal

reference voltage.

2. The system of claim 1, wherein: the calibration process is configured to calibrate the IB associated with the command-address (CA) circuit path; the IB and the logic comprise a device; a tester coupled to the device and configured to provide the external reference voltage and manage the calibration process simultaneously for two or more devices-under-test (DUTs), wherein the two or more DUTs include the device.
3. The system of claim 2, wherein: each of the two or more DUTs include a plurality of CA pads, wherein each CA pad is connected to a corresponding IB; and the tester is configured to provide the external reference voltage and manage the calibration process simultaneously for the plurality of CA pads and the corresponding IBs.
4. The system of claim 1, further comprising: a tester coupled to the IB and configured to provide the external reference voltage and manage the calibration process; wherein: the IB and the logic comprise a memory device coupled to the tester for the calibration process; the calibration process includes at least a first transition and a second transition that change a controlling device between the tester and the memory device; the tester is configured to use (1) an asynchronous signal and (2) a bidirectional signal to control to coordinate the first and second transitions.
5. The system of claim 4, wherein: the asynchronous signal is an error reporting signal provided by the memory device; the bidirectional signal is a test data (TDQ) signal; the tester is configured to initiate the calibration process by setting a mode control signal, wherein the TDQ signal is controlled by the memory device at the initiation of the calibration process; the first transition is initiated by a transition in the error reporting signal after the initiation of the calibration process, wherein, following the first transition, the TDQ signal is controlled by the tester and the external reference voltage is provided by the tester; the second transition is initiated by the tester, wherein, following the second transition, the memory device (1) controls the TDQ signal, (2) captures a CA state based on the external reference voltage provided to the IB according to the adjustment of the TDQ signal, and (3) subsequently provides results associated with the captured CA state to the tester for determining the trim setting for the IB.
6. An apparatus, comprising: an input buffer (IB) configured to isolate from a load (1) a command-address (CA) signal during a normal operation according to an internal reference voltage and (2) an external reference voltage during a calibration process that determines the trim setting; a calibration logic coupled to the IB, the calibration logic configured to operate, during the calibration process, to (1) provide an external reference voltage to the IB and (2) isolate the IB from the internal reference voltage; and an IB calibration circuit coupled to the IB and configured to implement the calibration process at the apparatus using the external reference voltage.
7. The apparatus of claim 6, wherein: the calibration logic is configured to (1) receive a mode control signal for initiating the calibration process and operate an internal-isolation switch that is coupled between the IB and the internal reference voltage, wherein the calibration logic operates the internal-isolation switch according to the mode control signal to isolate the IB from the internal reference voltage according to the mode control signal.
8. The apparatus of claim 7, further comprising: a pad configured to receive (1) an operating signal during normal operation and (2) the external reference voltage during the calibration process; wherein: the IB includes a first differential input coupled to the pad and a second differential input coupled to the internal reference voltage through the internal-isolation switch; and the calibration logic includes a differential-connection switch connected across the first and second differential inputs and configured to operate complementarily with the internal-isolation switch for connecting both the first and second differential inputs to the external reference voltage according to the mode control signal.
9. The apparatus of claim 7, wherein the IB calibration circuit is configured to initiate local operations of the calibration process in response to the mode control signal, wherein the local operations of the calibration process includes (1) transitioning an asynchronous signal for

coordinating a control transfer to a tester and (2) suspending a clock signal.

10. The apparatus of claim 9, wherein: the IB calibration circuit is configured to receive a test data signal controlled by the tester following the transition of the asynchronous signal; and the IB calibration circuit is configured to receive the external reference voltage following the transition of the asynchronous signal.

11. The apparatus of claim 10, wherein the IB calibration circuit is configured to: identify a subsequent transition in the test data signal as implemented by the tester; in response to the subsequent transition, capture a result according to the external reference voltage processed by the IB; reestablish control over the test data signal following the subsequent transition; and restore CA operations for the IB instead of receiving the external reference voltage.

12. The apparatus of claim 6, further comprising: a plurality of pads each configured to receive (1) an operating signal during normal operation and (2) the external reference voltage during the calibration process; a set of IBs, including the IB, wherein each IB in the set thereof corresponds to one pad in the plurality of the pads; and the IB calibration circuit is configured to calibrate all IBs on the apparatus simultaneously using the external reference voltage.

13. The apparatus of claim 12, wherein the IB calibration circuit is configured to: generate a set of calibration results; capture the set of calibration results in parallel; convert the captured calibration results from parallel data streams to a serial sequence; and provide the captured set of calibration results in serial format using a bidirectional communication connection to a tester.

14. A method of calibrating an internal buffer (IB) configured to process operating signals during normal operations for a device, the method comprising: receiving an external reference voltage from a tester during a calibration process; providing the external reference voltage to the IB while isolating the IB from an internal reference voltage during the calibration process; and implementing the calibration process at the device using the external reference voltage.

15. The method of claim 14, wherein: the device includes multiple pads that each correspond to a unique IB and receive the external reference voltage for the calibration process; and implementing the calibration process at the device includes simultaneously calibrating the unique IBs for the multiple CA pads according to the external reference voltage.

16. The method of claim 15, further comprising: testing multiple devices simultaneously based on providing the external reference voltage in parallel to the multiple devices from the tester.

17. The method of claim 15, wherein implementing the calibration process at the device includes: capturing calibration results in parallel; converting the calibration results into a serial data sequence; and communicating the serially sequenced calibration results for communication to the tester and/or a system operator.

18. The method of claim 14, further comprising: receiving a mode control signal from a tester for initiating the calibration process, wherein implementing the calibration process at the device includes: isolating the IB from the internal reference voltage according to the mode control signal; facilitating connection of the external reference voltage to the IB; transitioning an asynchronous signal for coordinating a control transfer to the tester; and suspending a clock signal.

19. The method of claim 18, further comprising: receiving a test data signal controlled by the tester following the transition of the asynchronous signal; and receiving the external reference voltage following the transition of the asynchronous signal.

20. The method of claim 19, further comprising: identifying a subsequent transition in the test data signal implemented by the tester; in response to the subsequent transition, capturing a result according to the external reference voltage processed by the IB; reestablishing local control over the test data signal following the subsequent transition; and restoring operations for the IB instead of receiving the external reference voltage in response to the subsequent transition.
